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Effect of Mode of Operation on Gas Turbine Combustor Performance

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Technical Report Abstract Page

Title	Effect of Mode of Operation on Gas Turbine Combustor Performance		
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Abstract: The primary goal of this study is to quantify the differences in the Combustor Performance under different modes of operation as a function of the combustor exit temperature. The main focus is the impact on liner heat loading since liner durability is strongly dependent on the metal peak temperature as well as temperature gradients. The study also includes impact on emissions and acoustics. Three modes of operation are discussed: non premixed natural gas operation, Dry Liquid Fuel operation and Wet liquid Fuel. Limited premixed natural gas data are also discussed when appropriate. Experiments are conducted on a Single Nozzle Rig (SNR) at conditions representative of different power levels of F class Gas Turbines. The results are analyzed to show the natural gas vs. dry liquid fuel performance, followed by the effect of water injection and effect of load variation. The results show that heat loading impact is amplified in the head end region since this is where the flame stabilizes. The combustor head end average temperature is 100 degree higher for Dry Fuel Oil when compared to natural gas. However, the tail section of the liner is 50 degree lower for Fuel Oil. The difference is mainly attributed to the change in flame shape and flame radiation. CO emissions were roughly 80 ppm lower for Natural Gas. NOx is higher for Liquid Fuel (35-80% depending on flame temperature). Emissions profiles at the combustor exit are similar for both fuels with a variability of 40% indicating it is controlled by flow-flame interaction. Acoustics are lower in amplitude for non-premixed natural gas, but peak frequency is similar. Liquid fuel wet mode of operation results in significant reduction in the head end temperature due to the water impingement on the liner (~300F). However, the maximum temperature over the tail section is 50F higher than dry operation. Yet, the average tail temperature is similar under both modes of operation with higher variability under wet operation (70%). This strong temperature variability can influence the liner life. More than 80% reduction in NOx emissions is achieved with water injection. Water injection impact on CO is dictated by CO generation mode (quenching vs. dissociation). More important is the significant improvement in emissions profile at the combustor exit with water injection (60% reduction in variability). At higher load, the maximum liner temperature is 80-150 degree higher than part load. This is mainly attributed to the increase in air inlet temperature and increase in radiation heat flux which is dependent on both flame temperature and pressure.

More investigations are required to determine the impact on complicated engine configurations as well as to isolate the effect of flame shape.

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EFFECT OF MODE OF OPERATION ON GAS TURBINE COMBUSTOR PERFORMANCE

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ABSTRACT

The primary goal of this study is to quantify the differences in the Combustor Performance under different modes of operation as a function of the combustor exit temperature. The main focus is the impact on liner heat loading since liner durability is strongly dependent on the metal peak temperature as well as temperature gradients. The study also includes impact on emissions and acoustics. Three modes of operation are discussed: non premixed natural gas operation, Dry Liquid Fuel operation and Wet liquid Fuel. Limited premixed natural gas data are also discussed when appropriate. Experiments are conducted on a Single Nozzle Rig (SNR) at conditions representative of different power levels of F class Gas Turbines. The results are analyzed to show the natural gas vs. dry liquid fuel performance, followed by the effect of water injection and effect of load variation. The results show that heat loading impact is amplified in the head end region since this is where the flame stabilizes. The combustor head end average temperature is 100 degree higher for Dry Fuel Oil when compared to natural gas. However, the tail section of the liner is 50 degree lower for Fuel Oil. The difference is mainly attributed to the change in flame shape and flame radiation. CO emissions were roughly 80 ppm lower for Natural Gas. NOx is higher for Liquid Fuel (35-80% depending on flame temperature). Emissions profiles at the combustor exit are similar for both fuels with a variability of 40% indicating it is controlled by flow-flame interaction. Acoustics are lower in amplitude for non-premixed natural gas, but peak frequency is similar. Liquid fuel wet mode of operation results in significant reduction in the head end temperature due to the water impingement on the liner (~300F). However, the maximum temperature over the tail section is 50F higher than dry operation. Yet, the average tail temperature is similar under both modes of operation with higher variability under wet operation (70%). This strong temperature variability can influence the liner life. More than 80% reduction in NOx emissions is achieved with water injection. Water injection impact on CO is dictated by CO generation mode (quenching vs. dissociation). More important is the significant improvement in emissions profile at the combustor exit with water injection (60% reduction in variability). At higher load, the maximum liner temperature is 80-150 degree higher than part load. This is mainly attributed to the increase in air inlet temperature and increase in radiation heat flux which is dependent on both flame temperature and pressure. More investigations are required to determine the impact on complicated engine configurations as well as to isolate the effect of flame shape.

1 INTRODUCTION

Gas Turbine Combustor life is strongly dependent on liner durability. Liner durability is a function of two main parameters: maximum temperature and temperature gradients. In fact, temperature gradient is getting more focus with the current advancement in material properties. Liner temperature variability was shown to be strongly influenced by flow field

variation (for example: cup to cup variation) and TBC (Thermal Barrier Coating) thickness variation [1]. Bradshaw and Waitz showed that the life expectancy can be reduced by 20% because of such variations [1].

Another parameter that derives liner temperature variability is the mode of operation. Gas turbines are typically designed to have dual fuel capabilities (Gas and Liquid). The favorable mode of operation depends on customer needs which may be related to available resources and site environmental restrictions. The important question is how does the mode of operation influence the liner heat loading in terms of both the maximum metal temperature and liner temperature gradients?

Mode of operation doesnot only influence the liner heat loading, but also the combustion emissions and combustor operability (combustion dynamics). They all come in one package: "Combustor performance". Effect of wet vs. dry mode of operation on emissions is very well documented. Reduction of NOx concentration with up to 80% was reported [2, 3] and the concept was even demonstrated on aircraft engines (during takeoff only) [4]. Usually the water to fuel ratio by mass is around 1 for stable operation [5]. Water injection may significantly impact the combustor exit emissions profile [6]. Water injection is associated with several problems including: generation of instabilities, hardware deterioration, the need of water treatment system and increased system maintenance. However, to the best of the authors knowledge limited/ no published data is available to quantify the impact on heat loading.

This paper presents a study on the impact of mode of operation on liner heat loading, combustion emissions and combustion dynamics. Liner heat loading is studied with over 30 Thermocouples distributed over the liner backside. Combustion emissions are reported in terms of an average at the combustor exit as well as detailed profile via a traversable emissions probe. Three modes of operation are discussed: non-premixed natural gas operation, dry liquid fuel and wet liquid fuel. Non-premixed natural gas is selected as it offers a better baseline for comparison with dry liquid fuel (both operate in diffusion mode). Then, wet operation is chosen to study the impact of water injection. Also, the results are compared at higher P and T to study the effect of combustor loading on performance. Premixed natural gas data is also discussed when appropriate.

2 NOMENCLATURE

Avg	=	Average
CO	=	Carbon Monoxide
CO_2	=	Carbon dioxide
CRZ	=	Central Recirculation Zone
DistII	=	Number 2 heating oil (Distillate)
DLN	=	Dry Low NOx
F/A	=	Fuel to Air ration by mass
ft/s	=	Velocity in feet's/second
H_2	=	Hydrogen
Lb/s	=	mass flow in lb/s
ms	=	millisecond
Norm	=	Normalized
NO_x	=	Nitric Oxides
NO	=	Nitric Oxide
NO_2	=	Nitrogen Dioxide
O_2	=	Oxygen
P_3	=	Combustor inlet pressure
ppmvd	=	Particle per million volume dry
$\Delta P/P$	=	Percentage pressure drop across Combustor
R	=	Cylindrical liner radius
RQL	=	Rich Burn Quick Quench Lean Burn
SNR	=	Single Nozzle Rig
T_3	=	Combustor air/Fuel mixture inlet temperature
TC	=	Thermocouple
$T_{Combustor\ Exit}$	=	Equilibrium flame temperature based on F/A
Wt	=	Wet mode of operation on Liquid Fuel
X	=	represents species concentration

3 EXPERIMENTAL SETUP AND TEST CONDITIONS:

3.1 Experimental facility

The test setup is shown in Figure 1. Four fluid circuits are illustrated on the diagram. Air circuit is shown in red, water in black, natural gas in green and liquid fuel in blue.

Air circuit (red) comprises main air and secondary air. Main air provides both combustion and cooling air at a temperature of 700-875F and pressure of 135-225 Psia based on corresponding load and is fed through a four inch line to the 16inch vessel where the combustor is mounted. The corresponding main air flow rate ranges from ~ 6.5-12 lb/s. The main air is metered by an orifice flow meter with a typical accuracy of ~ 0.6%. Secondary air comprises atomization and purge air. Secondary air is fed from a dedicated high pressure compressor at a temperature of roughly 300 F through a 1 inch line. Atomization air and purge air are metered separately using an orifice and venturi meters respectively. Atomization air is required to generate a liquid fuel spray and improve liquid fuel combustion characteristics at part load operating conditions. Atomization air is delivered at a pressure ratio of ~ 1.3. The corresponding flow is ~ 0.07-0.1 lb/s based on operating conditions. Purge air feeds set of holes located on the nozzle tip for cooling and purging requirements as well as to protect the tip from coke build up under liquid fuel operation. The flow rate is 0.05-0.1 lb/s based on operating conditions. The figure also shows that atomization air is used for purging the liquid fuel cartridge and water circuit (only the circuit used for emissions control). The solenoids operate based on required operation scheme.

Water circuit (black) is fed from a high pressure pump and it is plumped such that water is used for exhaust quenching, emissions control and liquid fuel cartridge flush. Basically, liquid fuel cartridge flush can be done using water or air as mentioned in the previous paragraph.

Natural Gas circuit (green) is fed from a separate compressor. It is a typical pipeline natural gas having a methane content of approximately 96%. Gas composition in provided in Table 1. Gas is metered using a Coriolis type mass flow meter

with an accuracy of 0.5%. It is plumped to offer supply to the premixed as well as non-premixed circuits. The non-premixed natural gas circuit feeds the purge holes on the tip of the nozzle. In non- premixed operation mode, the purge air is isolated and only natural gas is flowing. Natural gas is then used both for tip cooling and as a fuel. Liquid Fuel Circuit (blue) is fed from a pumping skid which can provide up to 0.35 lb/s at 3000 pisa. Distillate number 2 is used as a typical gas turbine liquid fuel. The fuel composition is provided in Table 2**.** Liquid fuel is metered using a Coriolis mass flow meter. Again, the water line T's to the liquid fuel line to enable water flush capability.

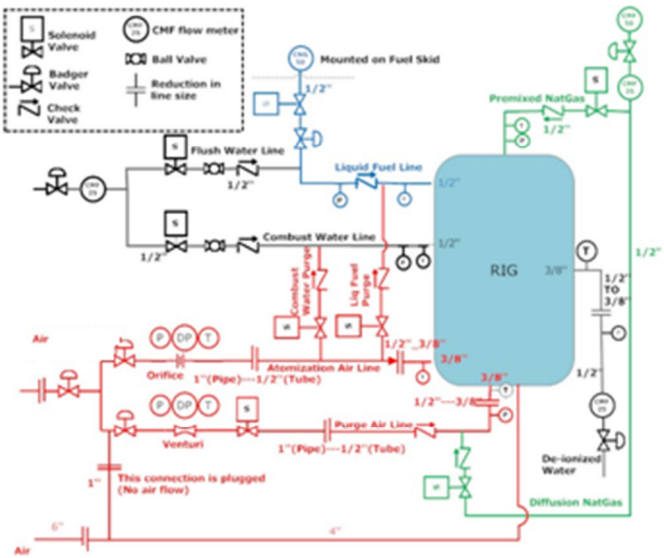

Figure 1 P&ID showing test setup.

Table 1 Natural Gas composition

Composition	Mole Fraction (%)
Methane	96.331
Ethane	1.543
Propane	0.215
Nitrogen	1.016
Carbon Dioxide	0.771

3.2 COMBUSTOR AND NOZZLE DESIGN

The combustor assembly is mounted on ASME stamped pressure vessels with diameter of 16" with pressure and temperature ratings of 370 psia and 1200°F respectively. The combustor assembly is illustrated in Figure 2. The air flow splits are illustrated in Figure 3. The overall air splits into three main portions. The main combustion air is roughly 79% of the total. The cooling air is roughly 21% of the total (backside cooling). Ninety four percent of the combustion air flows through the swirler or the nozzle and 6% is used for cap cooling. The atomization and purge air are separate circuits connected directly to the nozzle and not illustrated in the figure.

The combustion liner is shown in Figure 4. It is divided into two main zones. The primary zone has a 6.5" x 6.5" square cross section and is roughly ~10" long. The square cross section was selected to enable convenient optical access. The four corners of the square cross section have effusion holes to assist in cooling. The secondary zone (also referred to as Liner zone) has a circular cross section with a diameter of 6.5" and is ~ 25 inches long. It is chosen to be cylindrical for a more convenient cooling design. The reader shouldnot take the impression that any dilution or primary jets are involved. This is a DLN combustor and not an RQL type.

The nozzle is illustrated in Figure 5. It comprises a swirler, center body and a burner tube. The swirler spins the air to stabilize the flame via a CRZ as illustrated. The center body also aids in the process of flame stabilization because of the wake region that develops as the flow expands. Also, typically, corner recirculation zones exist and they also aid in flame stability. However, the flame stabilization is dependent on the combustion mode which dictates, based on fuel air mixing and flame velocity, where the flame will "sit". More discussion on aerodynamics and mode of operation will be presented in the results section. The burner tube just provides a space for air and fuel to mix in premixed mode of operation.

Table 2 Distillate No2. Composition

Properties	Distillate No.2
Sp. Gravity @ 60^0F	0.8576
API Gravity	33.5
Viscosity @ 100^0F (cSt)	2.690
Sulfur (% wt.)	0.111
Nitrogen (ppm weight)	290
Vanadium (ppm)	< 1
Carbon (% wt)	86.98
Hydrogen (% wt)	12.97
Wax Content (% wt)	< 5
LHV (BTU/lb)	18,225

3.3 Emissions Sampling

Combustion products are sampled using a water-cooled gas sampling probe placed on an axial traverse. The sample is then transferred to the sample-conditioning unit via heated line passing through a heated filter. A stream of the sample is then cooled, dried, and sent to a gas sampling system for CO_2, O_2, and CO measurement. The other stream remains heated before entering the NO_x analyzer to eliminate loss of NO_2. The gas sampling unit is calibrated using standard calibration gases. The

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analyzers are spanned prior to every test. In this report, NO_x and CO emissions are corrected and reported to 15% O_2 level by volume.

The emissions probe is mounted on a 1D traverse that enables scanning the emissions profile at the combustor exit, along the diameter, as illustrated in Figure 6. Assuming that any measured species is expressed as X, X is a function in the radius and can by averaged as given by Equation 1 where: X_{avg} is the average species concentration at the liner exit, R is the liner radius and r is a variable cylindrical coordinates. If the concentration is to be normalized then X_{Norm} can be estimated as given by Equation 2.

$$X_{avg} = \frac{\int_0^R X(r) 2(\pi)\, r dr}{\pi(R^2)} \qquad \text{Equation 1}$$

$$X_{Norm} = \frac{X(r)}{X_{avg}} \qquad \text{Equation 2}$$

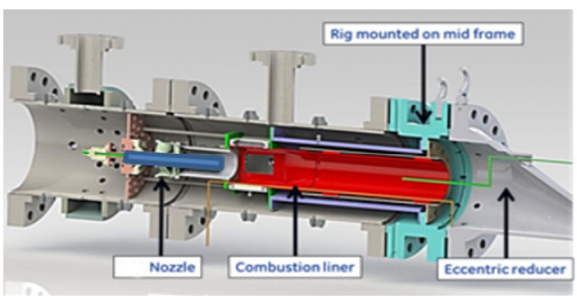

Figure 2 Combustor assembly in the Single Nozzle Rig.

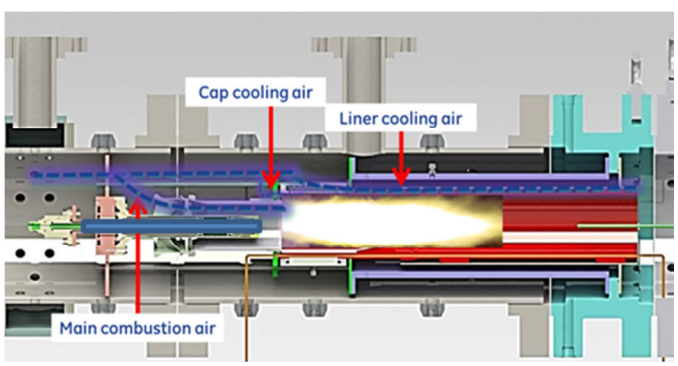

Figure 3 Combustor Air flow splits.

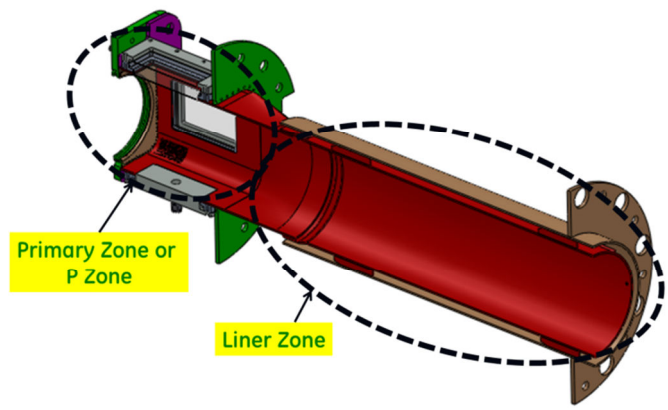

Figure 4 Combustion Liner

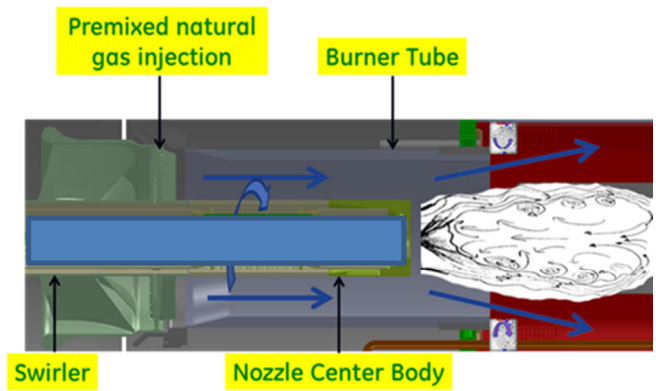

Figure 5 Nozzle illustration.

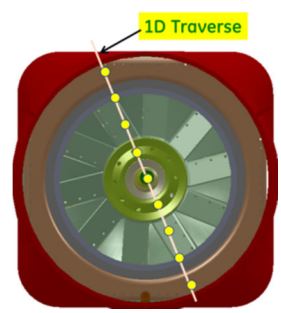

Figure 6 1D traverse capability illustration.

3.4 Test conditions and data acquisition

Experiments are conducted at P_3 of 140 (load 1) and 225 (load 2) psia and a corresponding T_3 of 715 and 865F. The combustor residence time is roughly 18 ms. The combustor pressure drop (ΔP/P3) varied from 2.5 to 4 % depending on operating conditions.

Real-time data is acquired at a frequency of 1Hz. Reported data points represent an average of 8 consecutive data samples. Maximum NO_x standard deviation is roughly ±1ppm. The reported average P_3 and T_3 has a typical standard deviation of ±1.2 psi (0.08 bar) and ±4°F (2.2 K) respectively. Flame temperatures are calculated using fuel-air ratios (F/A) based on

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metered air and fuel flows or the O_2 and/or CO_2 percentages in the sampled combustor products. The latter technique is valid when complete combustion occurs; i.e., at flame temperature above the LBO limits. In this study, the flame temperature based on O_2 and CO_2 typically match to within 30 degrees. In addition, the flame temperature based on metered air and fuel matches with the temperature based on O2 emissions within 1%. In this paper, reported flame temperatures are calculated based on F/A (equilibrium temperature). The uncertainty in flame temperatures is estimated to be ±40°F (22.2 K).

Figure 7 shows an illustration of the camera setup used to capture the flame shape. The camera is a regular surveillance camera that views the flame at a 90 degree angle through a 5.5" x 3" rectangular Quartz window.

4 RESULTS AND DISCUSSIONS

4.1 Natural Gas vs. Dry Liquid Fuel at part load operating conditions

4.1.1 Liner Heat loading

Figure 8 shows a plot of the maximum metal skin temperature vs. the combustor exit temperature for non-premixed natural gas (squares) and dry DistII (stars). The blue and the black lines represent the liner zone and primary zone respectively. Typically, the maximum metal temperature is the first concern to combustor designers when it comes to heat loading.

The first observation is the straight line relationship between the equilibrium flame temperature (combustor exit temperature) and the metal temperature. The overall heat transfer is dictated by the flame temperature, for a fixed cooling flow rate. The slope of that straight line is a function in the flame shape, the fuel properties and cooling flow rate.

Figure 8 shows that the behavior in the primary zone is very interesting. At flame temperatures up to 2300 F, maximum metal temperature is very similar for both fuels. However, as the flame temperature increases, the liner runs hotter with DistII and the difference reaches around 100 F at a flame temperature of ~2650F. At higher flame temperatures, both modes become similar again. This behavior in the primary zone is attributed to the flame shape.

Figure 9 shows the average temperature in the primary zone for both natural gas (dashed line) and dry liquid fuel (solid line). The average temperature is the average of 8 TC's located at two axial locations (4 TC's at each axial station). The error bars represent the standard deviation for all 8 readings. Basically, the standard deviation here reflects the variability in liner metal temperature in the primary zone. The results show that the primary zone runs roughly 100F higher, on average, under dry liquid fuel mode of operation. This is attributed both to the increase in radiation heat transfer (luminosity is higher for liquid fuel) and flame shape. The high variability in metal skin temperature for natural gas is aligned with difference in flame shape and stabilization for both fuels. Flame luminosity images support this conclusion.

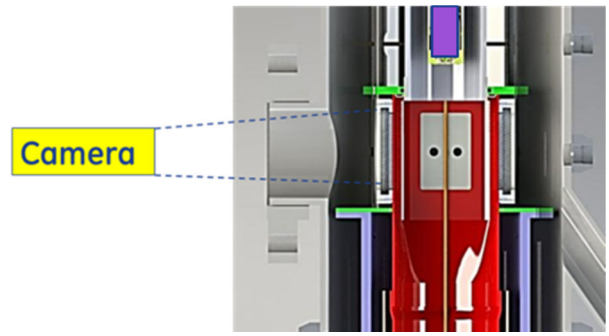

Figure 7 Camera setup

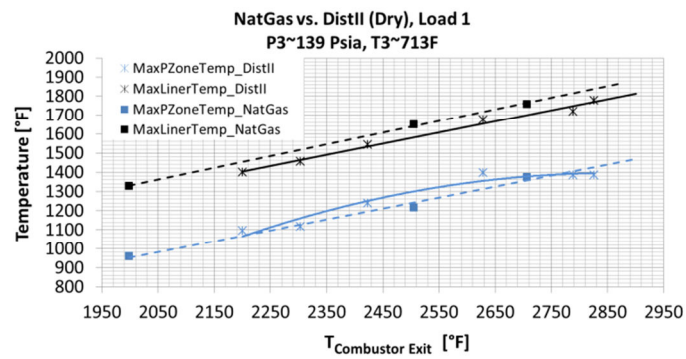

Figure 8 Maximum metal skin temperature vs. combustor exit temperature for non-premixed natural gas and dry Dist II at part load conditions.

Figure 10 and Figure 11 show the flame shape for both non-premixed natural gas and dry liquid fuel modes respectively. The pictures show that the corner recirculation is more involved in flame stabilization with liquid fuel than in natural gas operation. This is why the metal temperature is higher for DistII. More on the relation between flow structure and flame location is given in [7]. Though premixed natural gas operation is not discussed in this paper, Figure 12 shows the flame shape under premixed mode of operation. It is clear that the Corner recirculation zone is significantly contributing to flame stabilization. Under this mode of operation the liner temperature in the primary zone is even higher than the reported values. Flame shape strongly influences the metal temperature in the combustor head end region which is the volume where flame stabilization takes place. It is worth mentioning that TC's in this area may be used to indicate that LBO is approached [8].

The black lines on Figure 8 show the maximum metal temperature in the liner zone for both fuels. The liner zone, illustrated in Figure 4, only sees a mixture of hot swirling gases generated from the primary zone with no much difference in shape. This is why the trends are liner trends for both fuels in this region. However, the maximum liner temperature is 50F higher for natural gas. It appears that this is primarily due to

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rotating hot streaks since the average temperature over the liner is similar for both fuels as will be discussed shortly.

Figure 13 shows the average metal skin temperature in the liner region for both modes as a function of flame equilibrium temperature. There is a perfect match in the average, but more variability is apparent with natural gas operation. Again, this is mainly attributed to the non-homogenous flame structure as shown in Figure 10. The swirling hot streaks affect individual TC's, yet the average behavior is very similar.

4.1.2 Combustion Emissions

Figure 14 presents a plot of the averaged combustion emissions (NOx and CO both corrected to 15% O2 in ppm volume dry) vs. flame temperature for both modes. The NOx emissions follow the well-known exponential trend because of the domination of the thermal mechanism. The CO generation mechanism is quenching at low temperatures and dissociation on the high side.

NOx and CO are higher for dry liquid fuel operation than natural gas operation as depicted in Figure 14. Carbon Monoxide (CO) emissions from natural gas is roughly 85 ppm lower over the temperature range of interest. NOx emissions are 35-80% lower for natural gas over the range of interest with more reduction at higher flame temperatures. Though both fuels are running in a diffusion mode, yet there is a considerable reduction in emissions as we go from liquid to gas. This behavior is attributed to the vaporization of the liquid fuel spray as it burns (droplet burning mode). This mechanism dominates and dictates the emissions generated from the dry liquid fuel mode of operation. Though the premixed natural gas emissions are not discussed here, but under premixed natural gas mode of operation the NOx emissions is reduced by ~ 95% when compared to the non-premixed natural gas mode of operation.

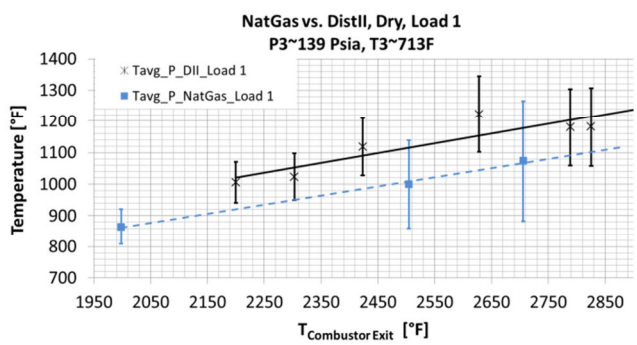

Figure 9 Average metal skin temperature in the primary zone vs. combustor exit temperature for non-premixed natural gas and dry Dist II at part load conditions.

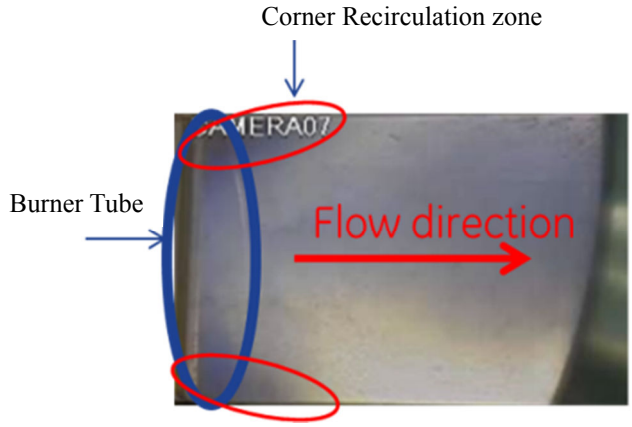

Figure 10 Non-Premixed Natural as Flame Shape

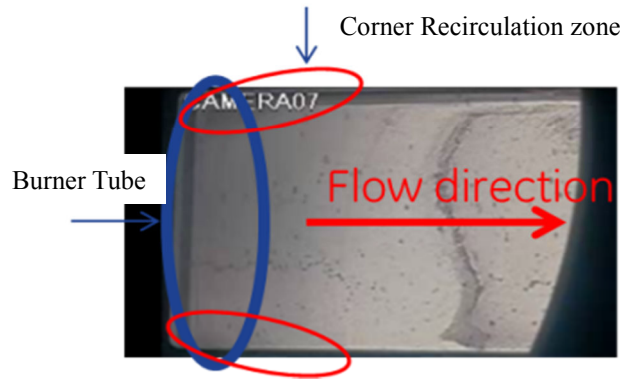

Figure 11 DistII (dry) flame shape

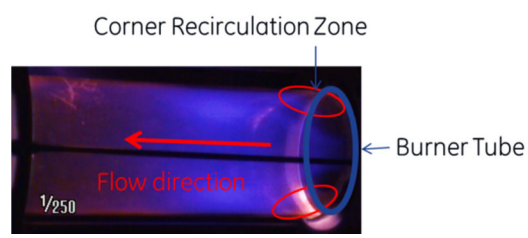

Figure 12 Premixed flame shape

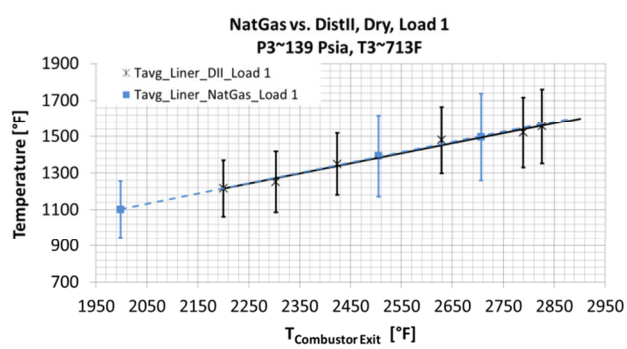

Figure 13 Average metal skin temperature in the liner zone vs. combustor exit temperature for non-premixed natural gas and dry Dist II at part load conditions.

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More important is the emissions profile at the combustor exit since it dictates the thermal loading on the first stage nozzle. Figure 15 shows the normalized O2 emissions over the combustor exit cross section for non-premixed natural gas mode. The values are normalized by the cross section average as given by Eq. (2) (the y-axis represents a percentage from the average). The variability is ~ 40% from the walls to the center point. The importance of O_2 profile is that it translates into a temperature profile as shown in Figure 16. Operation on dry liquid fuel results in similar variability as shown in Figure 17. This indicates that the variability is mainly due to the flow-flame interaction (this is a diffusion flame mode). However, as the combustor exit temperature exceeds 2700F, the variability increases to 70%.

Figure 18 and Figure 19 present the normalized NOx emissions profiles for non-premixed natural gas and dry liquid fuel respectively. Variability is around 170% for both modes though from an absolute perspective the NOx emissions are much lower for non-premixed natural gas (see Figure 14). This indicates that generation zones in both modes are very similar (for example CRZ as shown in Ref. [9, 10]). It is really interesting that this is the case though the combustor doesnot run with any primary or dilution jets (DLN design). A similar variability in an RQL type of combustor running on natural gas was reported in Ref. [6]. This indicates the importance of determination of emissions profiles or area weighted average emissions for nay combustor arrangement even if it is a DLN design. It is also note-worthy that the O_2 profile goes hand in hand with the NO_x profile (NO_x increases as O_2 decreases).

Figure 20 shows the CO emissions profiles for non-premixed mode of operation. At low temperatures CO originates from the wall due to a quenching mechanism. At higher temperatures CO is generated due to dissociation because of the high temperature of the center stream. Figure 16 shows that though the average exit temperature is 2700F, the stream at the center has a temperature that is close to 3200F. Dry liquid fuel shows similar CO trends and results are not shown here for reasons of brevity.

4.1.3 Combustor Operability

Combustor operability here refers to the combustion dynamics or pressure oscillations generated due to the coupling between heat release fluctuations and chamber acoustics. Figure 21 shows a plot of the peak frequencies and peak to peak (pk-pk) amplitudes vs. flame temperature for both modes of operation. Non-premixed natural gas amplitude of oscillation is within 1-2 psi pk-pk while the dry liquid fuel amplitude is within 5-7 psi pk-pk. The peak frequency for dry liquid fuel operation is ~ 220 Hz, but for natural gas varies between 150-220 Hz depending on flame temperature. As discussed previously, there is evidence that the corner recirculation zone involvement in heat release in the non- premixed natural gas mode is marginal when compared to Dry Liquid Fuel. This is mainly the reason the amplitudes of oscillations are significantly lower for the non-premixed natural gas mode. Interestingly, premixed natural gas mode shows low amplitude of oscillations when the flame stabilization mainly occurs in the CRZ. However, once the corner recirculation zone is more involved in the flame stabilization process, the amplitude of oscillations increase exponentially. More discussion on the regions of high velocity oscillations and their role in generation of thermo-acoustic instabilities is given in Refs. [11, 12]. Typically, combustion dynamics is suppressed in the engine by either staging or pilot operation.

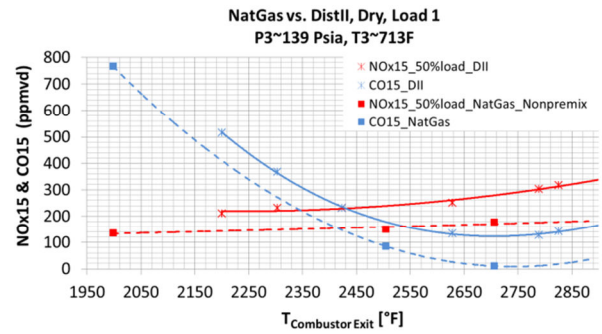

Figure 14 Average NOx15 and CO15 vs. combustor exit temperature for non-premixed natural gas and dry Dist II at part load conditions.

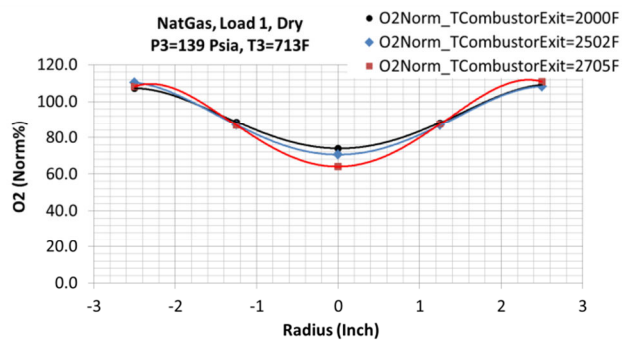

Figure 15 Normalized O2 emissions profiles over the combustor exit for non-premixed natural gas operation.

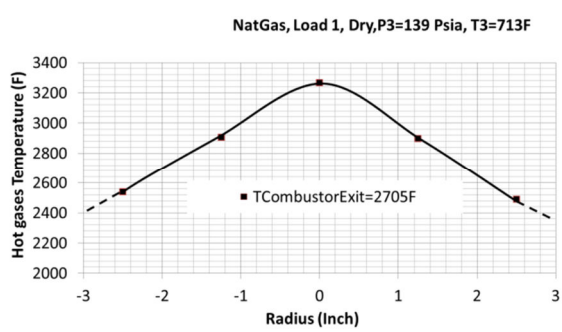

Figure 16 Temperature profile over the combustor exit for non-premixed natural gas operation at an average exit temperature of 2700F.

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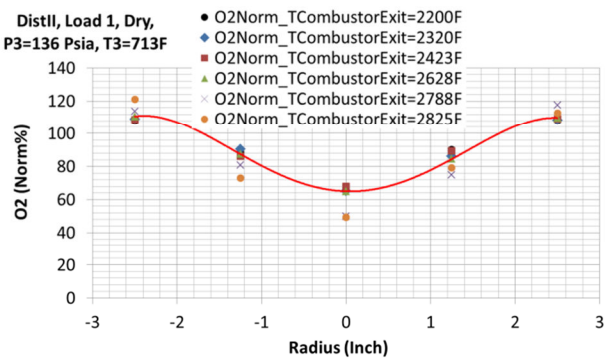

Figure 17 Normalized O2 emissions profiles over the combustor exit for Dry Liquid Fuel operation.

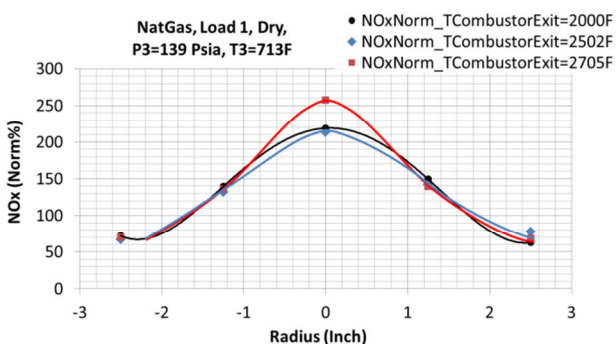

Figure 18 Normalized NOx emissions profiles over the combustor exit for non-premixed natural gas operation

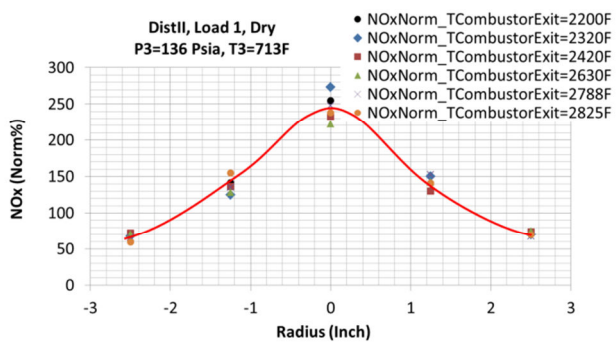

Figure 19 Normalized NOx emissions profiles over the combustor exit for Dry Liquid Fuel operation.

4.2 Dry vs. wet liquid fuel mode of operation

4.2.1 Liner Heat loading

Figure 22 shows a plot of the maximum metal temperature in the primary and liner zones under wet and dry liquid fuel modes of operation. The primary zone maximum temperature is 350 F lower with water injection, on average, due to the water impingement on the metal. The maximum liner zone temperature is roughly 50F higher with water injection. This is probably due to the increase in flame length that typically associates water injection. The rate of increase of metal temperature with increase combustor exit temperature is similar for both modes in the liner zone. However, it is much different in the primary zone. This is where the water injection has the strongest effect of flame stabilization and flame shape and therefore the response is totally different.

From an average perspective, the liner zone temperature is similar with and without water injection as depicted in Figure 23. However, the temperature variability with water injection is on average 70% higher. These variations results in higher temperature gradients under wet operation. Added to this is the effect of water on deterioration of hardware. This is why wet mode of operation is typically associated with more maintenance requirements. But again, the preferred mode of operation is mainly driven by emissions requirements as well as required pattern factor as will be discussed shortly.

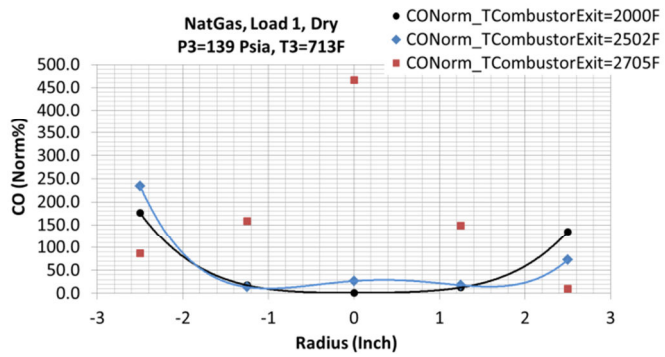

Figure 20 Normalized CO emissions profiles over the combustor exit for non-premixed natural gas operation.

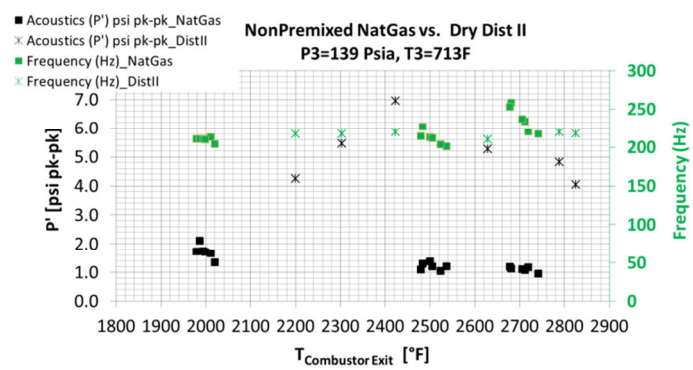

Figure 21 peak frequencies and corresponding amplitudes for non-premixed natural gas and dry DistII modes vs. combustor exit temperature.

4.2.2 Combustion Emissions

Figure 24 presents the average NOx and CO emissions (both corrected to 15% O_2) under wet and dry liquid fuel modes of operation. The merit of water injection is apparent with more that 80% reduction in NOx over a very wide range of combustor exit temperature. This behavior is mainly because water reduces the peak flame temperature and results in a more homogenous flame and thermal NOx are mainly a function of the peak temperatures.

Impact of water injection on CO depends mainly on the CO generation mode. At combustor exit temperatures less than 2600F, the generation mode is mainly quenching at the walls as discussed previously. Therefore, water injection at low

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temperatures results in increased CO emissions. On the contrary, at higher temperatures CO is generated due to dissociation from the central stream (center peaked profile). Consequently, water injection at combustor exit temperatures higher than 2650F results in decreased CO emissions. This agrees well with the results reported from an RQL system [6].

More of the water injection merits is manifested in the homogenization of the emissions profile at the combustor exit. The variability over the exit cross section drops from 40 to 20% as shown in Figure 25 (50% improvement). NOx emissions variability drops from 170% to 70% as shown in Figure 26 (60 percent reduction). CO profiles are not shown for reasons of brevity.

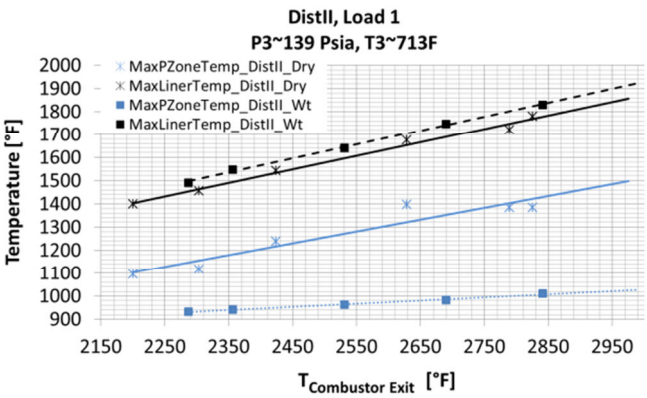

Figure 22 Maximum metal skin temperature vs. combustor exit temperature (dry & wet liquid fuel operation at part load).

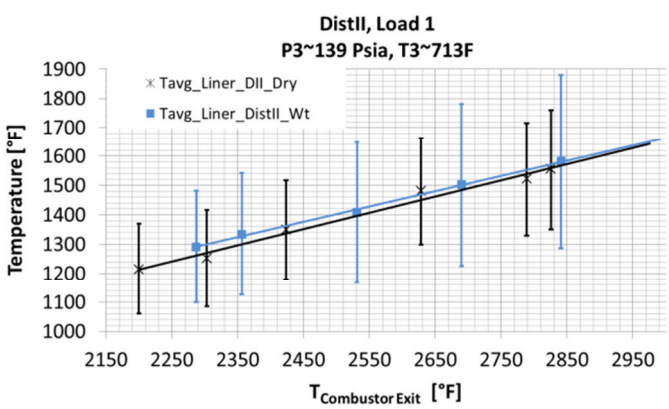

Figure 23 Average metal skin temperature in the liner zone vs. combustor exit temperature for dry and wet liquid fuel operation at part load conditions.

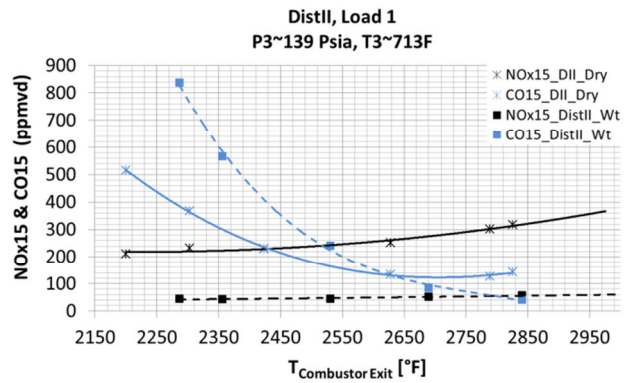

Figure 24 Average NOx15 and CO15 vs. combustor exit temperature for dry and wet liquid fuel operation at part load conditions.

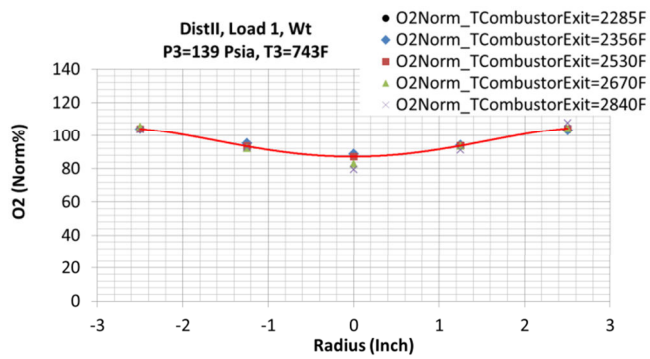

Figure 25 Normalized O2 emissions profiles over the combustor exit for wet liquid fuel mode of operation.

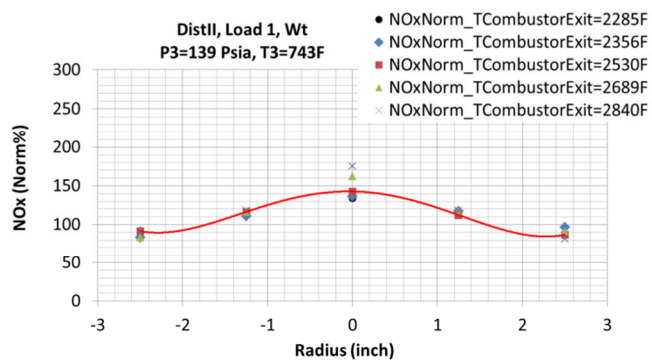

Figure 26 Normalized NOx emissions profiles over the combustor exit for wet liquid fuel mode of operation.

4.2.3 Combustor Operability

Figure 27 presents the effect of water injection on combustion dynamics. The amplitude of oscillations is roughly 2 psi lower with water injection, on average. The peak frequency is ~ 225 Hz for both modes. The impact observed here is not a universal impact. In fact, water injection may result in amplification of the amplitude of oscillations. For example, operation of the same combustor arrangement with higher pressure drop resulted in 5X increase in acoustics levels

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with water injection, not shown here. As the pressure drop increases, the turbulence activities increase while the flow maintain the same structure [7, 13]. However, the increase in turbulence activities results in propagation of the heat release into a larger portion of corner recirculation zone [14] or any new zone where the flame was not initially anchored. Once the water is turned on, such regions become the main source of instabilities. This is just a possible explanation to the observed phenomenon but more investigations are required to confirm the relationship between pressure drop and sensitivity to water injection as well as the combustion mode.

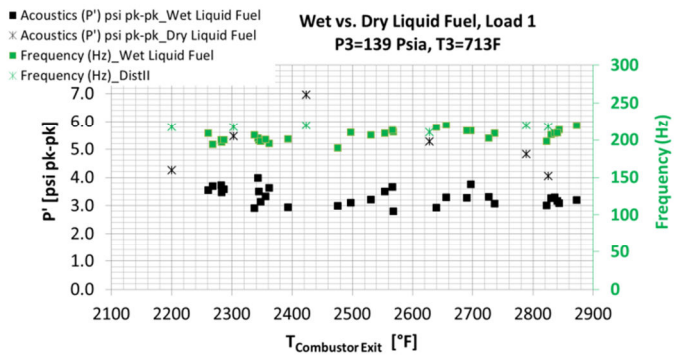

Figure 27 peak frequencies and corresponding amplitudes for dry and wet Liquid fuel modes vs. combustor exit temperature.

4.3 Full load vs. part load

4.3.1 Liner Heat loading

Figure 28 shows a plot of the maximum metal temperature in the primary zone (blue curves) and liner zone (black curves) at part (solid lines) and full load (dotted lines) operating conditions.

The primary zone maximum metal temperature is 280 F to 360 F higher at full load operating conditions. In other words, the lines fitting the data are not parallel but diverge as temperature increases. The higher heat loading is attributed to two factors. First, the change in flame shape due to the density differences and effect of T3. Second, the radiation heat transfer which is a function in luminosity which is a function in combustor pressure. It is well known that as the pressure increases flame radiation becomes more important [5, 15] so basically more heat is delivered to the liner via radiation mechanism. As more fuel is burned, more carbon particles are available to radiate heat and this derives the divergence of the lines with increased temperature.

The liner zone maximum metal temperature increases by 80F to 150F depending on the flame temperature. It is very interesting that the major impact on the liner at high power conditions occurs in the primary zone where flame shape strongly influences the heat transfer process. The metal temperature in the liner zone is ~130F higher at high power conditions when compared to part load from an average perspective (Figure 29). It is interesting that T_3 increased by ~ 130F from part load to higher load operating conditions. This reflects the important impact of T_3 which appears clearly when you look from an average perspective. Variability of temperature along the liner is very similar for both loads as depicted in Figure 29.

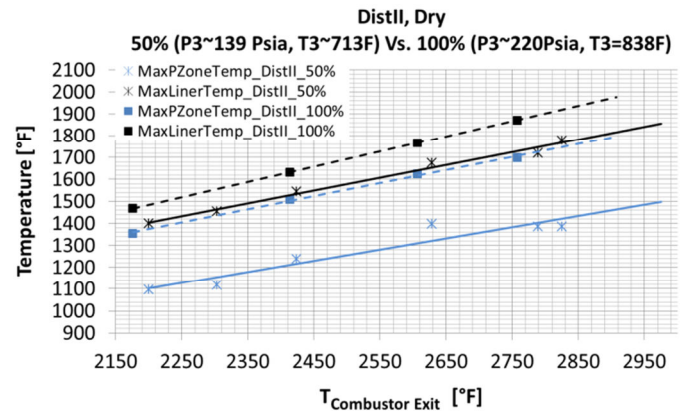

Figure 28 Maximum metal skin temperature vs. combustor exit temperature for dry mode of operation at low power and high power operating conditions.

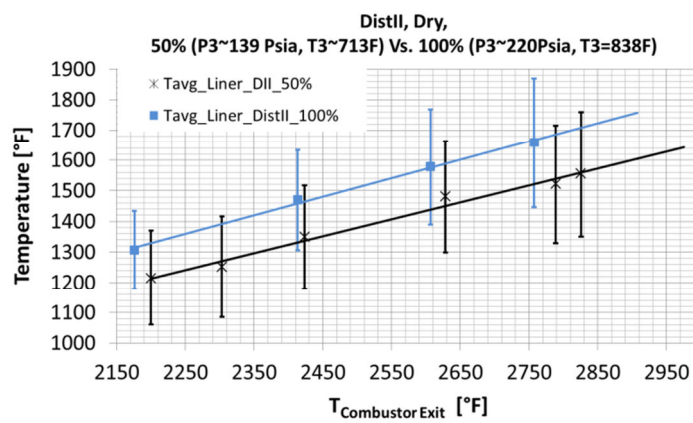

Figure 29 Average metal skin temperature in the liner zone vs. combustor exit temperature for low power and high power operating conditions.

4.3.2 Combustion Emissions

Figure 30 presents the average NOx and CO emissions (both corrected to 15% O_2) under low power and high power operating conditions versus combustor exit temperature. The high power conditions increases NOx by 55-70% depending on flame temperature. The effect is more pronounced as the flame temperature increases. The industrial standard for pressure effect is a 0.5 pressure exponent (or n) where NOx is proportional to P^n (P is the operating pressure). However, it has been shown that the pressure effect is a function in the flame temperature as well as pressure [5, 16, 17]. A plot of the pressure exponent derived from this data is given in Figure 31. The exponent varies from 0.9-1.05. This is not a pure pressure effect, but also includes a T_3 effect and is based on a diffusion flame mode.

CO is reduced by 84-66% depending on flame temperature. CO suppression is mainly due to the reduction in dissociation reactions as the pressure increases. Also, the wall quenching mechanism is reduced since the liner runs hotter as discussed previously.

4.3.3 Combustor Operability

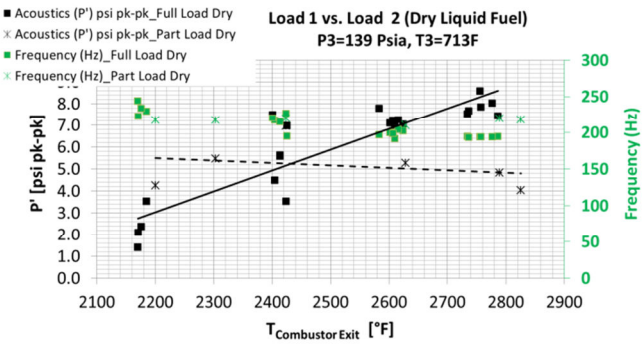

Figure 32 shows peak frequencies and amplitudes of oscillations at low power and high power. The behavior is a typical behavior in single nozzle rigs where the amplitude of oscillations is amplified as the operating pressure increases [18]. Peak frequencies are still close under both conditions (~ 200Hz). Dynamics is provoked by the presence of heat release oscillations. As the operating pressure increases more fine structures are present in the flow and turbulence intensifies. This is the suitable environment for oscillations to amplify. It appears that this is why as the temperature (or fuel flow) is increased the amplitude of oscillations increases at high power conditions. From an engine perspective, the choked first stage nozzle typically reduces reflectivity and the dynamics level are much lower than SNR's.

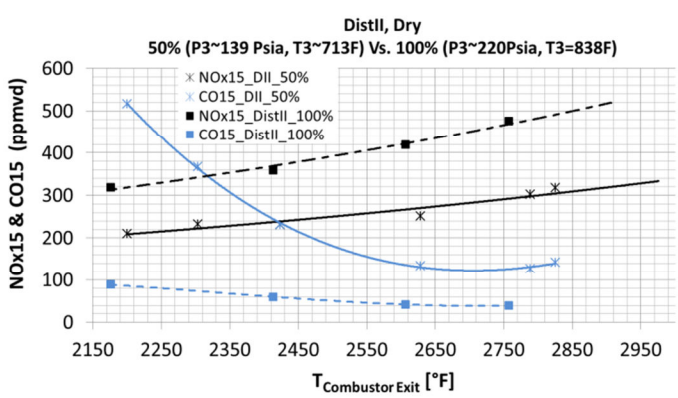

Figure 30 Average NOx15 and CO15 vs. combustor exit temperature for dry liquid fuel operation at low power and high power operating conditions.

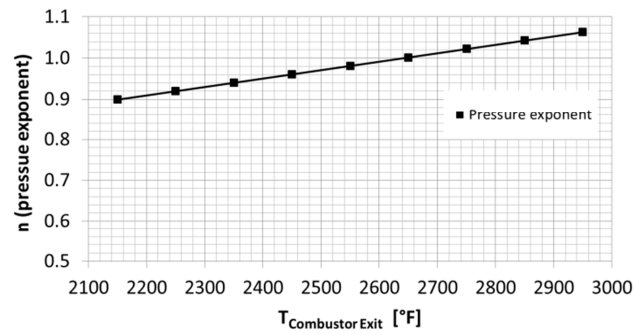

Figure 31 pressure exponent n as function of flame temperature.

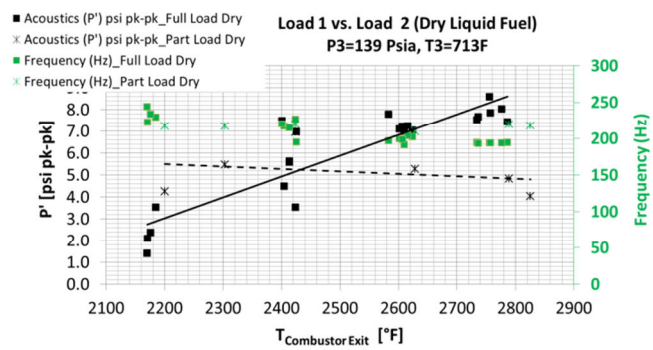

Figure 32 peak frequencies and corresponding amplitudes for low power and high power operating conditions vs. combustor exit temperature.

5 CONCLUSIONS

Mode of operation strongly influences liner heat loading. The impact of mode of operation is amplified at the head end region or the primary region where flame stabilization takes place. This is true for the combustor even if it is configured with backside cooling and with no primary or dilution holes (DLN design). The impact of mode and/or fuel on the tail section of the liner is relatively lower than the head end region.

Switching from natural gas to dry liquid fuel resulted in 100F increase in the average head end temperature but with 50% decrease in temperature variability due to a more homogenous flame structure. The tail section was 50F lower for dry liquid fuel.

Turning the water on, for liquid fuel, resulted in 300F reduction in the head end temperature with 50% increase in temperature variability mainly due to water impingement on the metal. The tail section of the liner was running 100F higher with water injection, mainly due to the increase in flame length that associates water injection.

Increasing the power level resulted in the head end running 280-360F higher than part load. The tails section average

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temperature was roughly 130F higher which confirms to the increase in T_3.

Emissions on natural gas are lower than dry liquid fuel though both run in diffusion flame mode. NOx is 35%-80% lower and CO is 80 ppm lower. Difference is mainly due to the droplet burning mode in liquid fuel. Interestingly, emissions profiles are similar for both with 40% variability in the radial direction. This indicates that the profile it is dictated by flow-flame interaction. Turning the water on results in 50% improvement in the profile variability along the radius. NOx is reduced by more than 80%. Increase in power level results in NOx increase by a pressure exponent of ~0.95 and 75% reduction in CO.

Pressure oscillations amplitude (combustion dynamics) are lowest for non-premixed natural gas followed by wet liquid fuel, dry liquid fuel and worst at higher load conditions.

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